

Pools, Rules, and Tools: A Template-Integrated Resource Architecture

Pools, Rules, and Tools: A Template-Integrated Resource Architecture I

**A Meta-Project Demonstrating Fonds, Rules, and Tools
Integration in Research Software**

Field	Value
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Author Contributions I

Daniel Ari Friedman: Conceptualisation, architecture design, module implementation (`fonds_reader`, `rules_applier`, `tools_invoker`, `integration`, `strong_rule_evaluator`, `figures`, `manuscript_variables`), manuscript writing (all sections), test suite design, exemplar resource authoring (`fonds`, `rules`, `tools`), validation.

Acknowledgements I

The author thanks the Active Inference Institute for hosting the public template repository and providing the infrastructure within which this exemplar was developed. The design of the three-layer resource architecture draws inspiration from the Unix philosophy [Raymond2003art] and the enterprise application patterns documented by [Fowler2002patterns].

Data Availability I

All source code, configuration files, and exemplar resources described in this paper are available in the public template repository under the `projects/templates/template_pools_rules_tools/` path. The integration pipeline is fully reproducible from source using `uv run python projects/templates/template_pools_rules_tools/scripts/02_run_integration.py` from the repository root. Generated manuscript variables are stored in `output/data/manuscript_variables.json` and injected at render time. Every figure in this manuscript, including the cover illustration, is likewise reproducible from source via `uv run python projects/templates/template_pools_rules_tools/scripts/05_generate_figures.py`, which writes directly to `manuscript/figures/`.

Competing Interests I

The author declares no competing interests.

Reproducibility Checklist I

This exemplar targets full computational reproducibility for every quantitative claim and figure it makes:

Artefact	Regenerated by	Verified by
Integration counts (fonds/rules/tools)	scripts/02_run_integration.py	tests/test_integration.py end-to-end assertion
Manuscript variable tokens	scripts/03_generate_manuscript.py	Rendered PDF contains no unresolved {{...}} tokens
All 9 figures (8 content + 1 cover)	scripts/05_generate_figures.py	tests/test_figures.py (per-function file-existence checks)
90% src/ line coverage	uv run pytest ... --cov-fail-under=90	CI + local pre-push gate
Combined PDF	full project pipeline, Stage 6 (4-pass xelatex + bibtex)	pdftotext scan for unresolved-reference markers and page-scale raster read

Reproducibility Checklist II

A reader who clones the repository and runs the five commands above, in order, reproduces every number and image in this document without manual intervention. Every quantitative claim in this document — fond/rule/tool counts, test counts, coverage percentages, and both bar-chart figures (@fig:counts, @fig:pipeline) — is generated from the same `run_integration_demo()` call at render time. None of these numbers are hand-typed; a `test_reflects_changed_integration_result` negative control in `tests/test_manuscript_variables.py` proves the token-generation function actually tracks its source rather than emitting a fixed constant.